

1- Data in the real world is dirty. Discuss this statement

Real-world data is considered "dirty" because it is often filled with flaws such as faulty instrument readings, human/computer errors, or transmission problems. Specifically, the data can be **incomplete** (missing values or attributes), **noisy** (containing errors or outliers), and **inconsistent** (having discrepancies in codes or naming conventions).

2- What is the difference between symmetric and Asymmetric binary data types

- **Symmetric binary:** Both outcomes have equal importance (e.g., gender).
- **Asymmetric binary:** The outcomes are not equally important (e.g., a medical test result like HIV positive vs. negative), and the convention is to assign the value "1" to the most important outcome.

3- Define the main Apriori principle and the two main steps this Algorithm

- **Apriori Principle:** If an itemset is considered frequent, then all of its subsets must also be frequent.
- **Two Main Steps:**
 - **Frequent Itemset Generation:** Find all itemsets whose support is greater than or equal to the minimum support threshold.
 - **Rule Generation:** Generate high-confidence association rules from those frequent itemsets.

4- List the Knowledge Discovery (KDD) Process steps.

The steps are: Data preparation, Data cleaning, Data integration, Data transformation, Data selection, Data mining, Pattern/model evaluation, and Knowledge presentation.

5- List different qualitative data types with example

- **Nominal:** Categories or "names of things" without a specific order (e.g., hair color, marital status, or zip codes).
- **Binary:** A nominal attribute with only two states (e.g., true/false or gender).
- **Ordinal:** Values that have a meaningful order or ranking, but the exact difference between them is unknown (e.g., sizes like small, medium, large).

6- List different Data cleaning tasks

- Identify outliers and smooth out noisy data.
- Fill in missing values.
- Correct inconsistent data.
- Resolve redundancy caused by data integration.

7- There are various reasons for missing values. Discuss how to handle missing values

Missing data can be handled using several methods:

- **Ignore the tuple:** Often done when the class label is missing.
- **Fill in manually:** Tedious, but an option.
- **Fill in automatically using:** A global constant (like "unknown"), the attribute's mean, the attribute mean for all samples belonging to the same class, or the most probable value (using Bayesian formulas or decision trees).

8- List three data types with example

- **Record Data:** Relational tables or transaction data.
- **Graphs and Networks:** Transportation networks, the World Wide Web, or social networks.
- **Ordered Data:** Video data (sequences of images) or temporal/time-series data.

9- Distinguish between noise and outliers.

- **Noise:** A random error or variance in a measured variable.
- **Outlier:** A data object that does not comply with the general behavior of the data and has characteristics that are considerably different from most other data objects.

10- What is the Major Tasks in Data Preprocessing

- **Data cleaning:** Handling missing data, smoothing noisy data, and removing outliers.
- **Data integration:** Combining multiple databases, data cubes, or files.
- **Data reduction:** Dimensionality reduction, numerosity reduction, and data compression.
- **Data transformation and discretization:** Tasks like normalization and concept hierarchy generation.